

Adil Karjauv

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EDUCATION

• KAIST, Robotics and Computer Vision (RCV) Lab

Daejeon, Republic of Korea

Master of Science in Electrical Engineering

Sep. 2019 – Jun. 2021

• KAIST

Daejeon, Republic of Korea

Bachelor of Science in Computer Science

Sep. 2014 – Feb. 2019

EXPERIENCE

• Qualcomm AI Research

Amsterdam, Netherlands

Machine Learning Researcher (Senior Engineer)

Oct. 2022 – Present

- Conducting research on efficient video processing and video generative AI:
 - **MobileWan:** Led an architecture optimization effort for the largest video generation diffusion transformer running on-device (based on Wan2.2-5B), developed a novel structured pruning method, improving efficiency without sacrificing quality. Paper under review.
 - **PyramidalWan:** Co-developed a step distillation method for a pyramidally-distilled video generation model, achieving a 10x inference speedup. Published at CVPR 2026; patent filed.
 - **Neodragon:** Contributed to step distillation techniques (2.5-10x speedup) for the first diffusion transformer based mobile video generation model. Presented as a NeurIPS 2025 demo and published at ICLR 2026; patent to be filed.
 - **MoViE:** Developed and optimized a state-of-the-art diffusion model for video editing, achieving an 87x on-device speedup – the fastest diffusion-based video editing demo shown on-device. Presented at NeurIPS 2024 and CVPR 2025 as a demo.
 - **Object-Centric Diffusion (OCD):** Contributed to training-free techniques that reallocate compute toward salient regions in video editing, cutting latency up to 10x with comparable quality. Published at ECCV 2024.
 - Pioneered the first on-device deep-learning-based video denoising solution at high resolution (QHD/4K at 30FPS); patent filed.

Technology Stack: Python, PyTorch, DeepSpeed.

• Robotics and Computer Vision (RCV) Lab

Daejeon, Republic of Korea

Graduate Student Researcher

Sep. 2019 – Jan. 2022

- Conducted research on machine learning and computer vision with a focus on adversarial machine learning and its applications in multimedia:
 - Improved hiding capacity of robust deep hiding by 30x over the state-of-the-art.
 - Increased data-free universal attack fooling ratio from 73.59% (state-of-the-art) to 84.08% in the white-box scenario and transferability of the attack from 39.05% (state-of-the-art) to 56.96%.

Technology Stack: Python, PyTorch, TensorFlow, NumPy, OpenCV.

SELECTED PUBLICATIONS

- Mohsen Ghafoorian*, Denis Korzhenkov*, **Adil Karjauv***, ... (*Equal contribution), “MobileWan: Closing the Quality Gap for Mobile Video Diffusion”, **arXiv preprint**.
- Denis Korzhenkov*, **Adil Karjauv***, Animesh Karnewar, Mohsen Ghafoorian, Amirhossein Habibian (*Equal contribution), “PyramidalWan: On Making Pretrained Video Model Pyramidal for Efficient Inference”, in **CVPR 2026**.
- Animesh Karnewar, Denis Korzhenkov, Ioannis Lelekas, Noor Fathima, **Adil Karjauv**, ..., “Neodragon: Mobile Video Generation using Diffusion Transformer”, in **ICLR 2026**.
- **Adil Karjauv***, Noor Fathima*, Ioannis Lelekas, Fatih Porikli, Amir Ghodrati, Amirhossein Habibian (*Equal contribution), “MoViE: Mobile Diffusion for Video Editing”, **arXiv preprint**.
- Kumara Kahatapitiya, **Adil Karjauv**, Davide Abati, Fatih Porikli, Yuki M. Asano, Amirhossein Habibian, “Object-Centric Diffusion for Efficient Video Editing”, in **ECCV 2024**.

SKILLS AND TECH STACK

- **Programming languages:** Python, Java, C++, C, MATLAB.
- **Technologies:** PyTorch, DeepSpeed, Diffusers, TensorFlow, TensorBoard, Keras, NumPy, OpenCV, Git, SQL, ROS.
- **Skills:** Deep Learning, Generative Modeling, Diffusion Models, Efficient ML (Distillation/Pruning), Distributed Training (DDP/FSDP/DeepSpeed), On-Device Inference, Multimodal & Foundation Models, Adversarial ML.